Aft-Fan Inflow Induction and Wing Separation During Tail-Sitter Transition

Onur Tuncer*

Istanbul Technical University, Istanbul, Türkiye

Çağlar Üçler†

Özyeğin University, Istanbul, Türkiye

Ahmet Güneş‡

Istanbul Technical University, Istanbul, Türkiye

A tail-sitter with an aft-mounted ducted fan spends its transition at incidences where the wing is separated, and the fan sits close enough behind the wing to change that. The mechanism is not the one a blown-wing study would model: the duct leading edge lies a fifth of a chord *aft* of the wing trailing edge, so its slipstream never reaches the wing. What reaches the wing is the fan's *upstream induction* — the accelerating field ahead of a loaded disk — which is a favourable streamwise pressure gradient and therefore delays separation. This paper models that induction as the semi-infinite cylindrical vortex sheet equivalent to a uniformly loaded actuator disk, verified against the closed-form axial solution and against momentum theory, and applies it as an external field to a coupled wing–body panel solve whose separation behaviour is established independently. On the demonstrator airframe the fan is found to buy $\Delta\alpha \approx 0.8^\circ$ of incidence at the transition point, measured as the shift of the separation-location curve rather than as the crossing of a single threshold — a distinction that matters, since a threshold evaluated on a two-degree grid reports no effect at

*Professor, Department of Aeronautical Engineering; onur.tuncer@itu.edu.tr. TODO: AIAA member grade.

†Associate Professor; caglar.ucler@ozyegin.edu.tr. TODO: department and AIAA member grade.

‡Associate Professor, Faculty of TODO; TODO: email and AIAA member grade.

all from data containing this one. The benefit is concentrated inboard, over precisely the span station that separates first, and reverses sign outboard where the wing leaves the fan's streamtube and meets its return flow.

Nomenclature

A	disk area, m^2
b	wing span, m
c_n	chord in the leading-edge-normal plane, m
C_L	lift coefficient
R	disk radius, m
s	axial distance from the disk plane, m
T	fan thrust, N
u_x	axial induced velocity, m/s
v_h	hover induced velocity, $\sqrt{T/(2\rho A)}$, m/s
v_i	induced velocity at the disk plane, m/s
V_∞	freestream speed, m/s
x_{sep}	chordwise separation station, fraction of c_n
α	angle of attack, deg
γ_t	cylindrical vortex sheet strength, m/s
μ	velocity ratio, V_∞/v_h
ρ	density, kg/m^3

I. Introduction

Transition is the part of a tail-sitter's envelope that sizes its wing and its control authority,[1,2] and it is the part a potential-flow method is least able to speak to, because the wing is separated over much of it. The companion study of this airframe[3] establishes where separation begins and how far forward it moves with incidence, marching an integral boundary layer from the located attachment point.[4,5] This paper asks what the propulsion system does about it.

The question is usually posed for a *tractor* installation, where the propeller slipstream washes the wing and raises its dynamic pressure — a blown wing, studied exhaustively for both conventional and wingtip-mounted layouts.[6,7] The airframe here is the other arrangement, and the difference is not a detail. Its ducted fan — a configuration whose static performance and lip-suction thrust split are classical[8,9] and whose small-scale behaviour is well characterized[10] — sits at the tail: the duct leading edge is 0.21 chord *behind* the wing trailing edge, so the slipstream leaves downstream and never touches the wing. A model that represents only the wake therefore reports exactly zero interaction, which is a statement about the model rather than about the aircraft.

What does reach the wing is the induction *ahead* of the disk. A loaded actuator disk accelerates the flow approaching it, from zero far upstream to v_i at the disk plane;[11,12] a wing sitting in that field experiences a favourable streamwise pressure gradient, and a favourable gradient delays separation.[13] The effect is the same in sign as a blown wing's and entirely different in mechanism. Its closest studied analogue is the upstream blockage of wind-turbine rotors, where the same vortex-cylinder model is used to quantify how far ahead of a rotor the flow is disturbed;[14] the present application reverses the question, asking what that disturbance does to a lifting surface placed inside it.

Figure effig:configuration shows the configuration and the field in question. Two features make it sharp rather than academic. First, the induced velocity is not a small correction: with the annular disk area of the demonstrator, v_h runs 14–24 m/s against a 25 m/s cruise, so the induction is comparable to the flight speed throughout transition and largest exactly where the airspeed is lowest. Second, the geometry aligns: the duct outer radius is 0.21 semi-span, and the companion study places the wing's most advanced separation at $|2y/b| \approx 0.15$–0.23. The part of the wing that sheds first is the part the fan protects.

II. The induction model

A. Why the wake model cannot serve

The solver's existing slipstream model reconstructs a rotor's wake from annular momentum theory[12] and returns zero induced velocity at any point ahead of the disk plane. That is a correct economy for a surface lying in the wake, and it is the wrong tool here by construction. Applied to a wing upstream of the disk it returns zero, and the zero is the model's rather than the flow's.

B. The disk as a vortex cylinder

A uniformly loaded actuator disk is exactly equivalent to a semi-infinite cylindrical vortex sheet.[15] The equivalence is worth deriving rather than asserting, because everything downstream of it — including the sign of the effect this paper reports — follows from the derivation.

Across a disk of radius R carrying uniform loading, the pressure jump Δp is constant over the disk face and the bound circulation on a blade element is therefore independent of

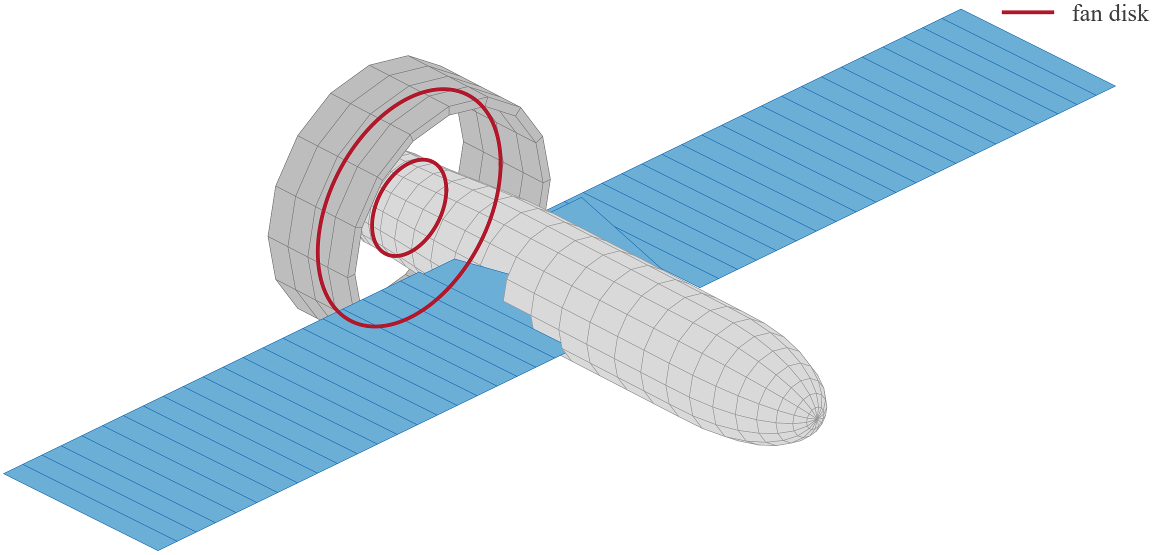

Figure 1. The coupled solve: the wing lattice, the source-panelled fuselage and duct, and the fan disk, all carried in one system. The fan sits behind the wing and inboard of its mid-span — the two geometric facts the rest of this paper turns on.

radius. Trailing vorticity is shed only where the bound circulation *changes*, so for uniform loading it is shed only at the edge $r = R$, and the shed sheet is convected downstream to form a cylinder. Its strength follows from the velocity jump across the sheet: far downstream the flow inside the streamtube is faster than outside by the fully developed wake increment, so

$$\gamma_t \;=\; [u_x]_{\text{out}}^{\text{in}} \;=\; 2v_i, \tag{1}$$

per unit axial length, with the vorticity purely azimuthal. This is the hydrodynamic analogue of a solenoid, and the axial field follows the same way.

The axis solution. Take the disk plane at $s = 0$ with the cylinder occupying $s \in [0, \infty)$. A ring element of the sheet at axial station s' carries circulation $\mathrm{d}\Gamma = \gamma_t\,\mathrm{d}s'$, and a circular vortex filament of radius R and circulation $\mathrm{d}\Gamma$ induces, on its own axis at a distance $\xi = s - s'$,

$$\mathrm{d}u_x = \frac{\mathrm{d}\Gamma}{2}\,\frac{R^2}{(R^2 + \xi^2)^{3/2}}. \tag{2}$$

Integrating Eq. (2) over the whole semi-infinite cylinder,

$$u_x(s) = \frac{\gamma_t R^2}{2} \int_0^\infty \frac{\mathrm{d}s'}{[R^2 + (s - s')^2]^{3/2}} = \frac{\gamma_t}{2} \left[1 + \frac{s}{\sqrt{s^2 + R^2}} \right], \tag{3}$$

which is the closed form the model is verified against. Its three limits are the three statements of momentum theory, recovered from the vortex system rather than assumed:

$$u_x(-\infty) = 0, \qquad u_x(0) = \frac{\gamma_t}{2} = v_i, \qquad u_x(+\infty) = \gamma_t = 2v_i. \tag{4}$$

The middle limit is the classical result that half the ultimate induction is present at the disk, and here it is a consequence of the cylinder being *semi*-infinite: the disk plane sees exactly half the sheet.

THE UPSTREAM FIELD, WHICH IS WHAT THIS PAPER NEEDS. Setting $s < 0$ in Eq. (3) gives the induction ahead of the disk, and expanding for $|s| \gg R$,

$$u_x(s) \;=\; \frac{\gamma_t}{2} \left[1 - \frac{|s|}{\sqrt{s^2 + R^2}} \right] \;\sim\; \frac{v_i R^2}{2s^2} \qquad (s \to -\infty), \tag{5}$$

so the upstream induction decays as the inverse square of distance — the signature of a semi-infinite vortex cylinder viewed from far upstream, where it appears as a point sink of strength $\pi R^2 \gamma_t$. The decay is fast, which is why the effect reported here is a near-field one and why the wing's proximity to the duct is essential rather than incidental. At one radius upstream Eq. (3) gives

$$u_x(-R) = \frac{\gamma_t}{2} \left(1 - \frac{1}{\sqrt{2}} \right) = 0.293\, v_i, \tag{6}$$

the number the whole interaction argument turns on, and the one Fig. 3 marks.

Off the axis the sheet is discretized into vortex rings, each polygonized into straight filaments and evaluated with the same Biot–Savart kernel the lattice uses for its own wake. Ring spacing grows geometrically downstream, since the first radius or two sets essentially the entire upstream answer while the far tail varies slowly, and that spacing keys to each cylinder's own radius rather than the disk's outer one.

THE ANNULUS. A ducted fan around a tail boom is an annulus, not a disk, and the same argument gives it directly. Uniform loading between R_h and R sheds trailing vorticity only where the bound circulation changes, which is now at *both* edges: $+\gamma_t$ at the outer radius and $-\gamma_t$ at the inner. The field is the superposition of two semi-infinite cylinders of opposite sign, so on the axis

$$u_x(s) = \frac{\gamma_t}{2}\left[\frac{s}{\sqrt{s^2+R^2}} - \frac{s}{\sqrt{s^2+R_h^2}}\right], \tag{7}$$

the constant terms of Eq. (3) having cancelled. Three consequences are worth reading off, because they are physical rather than numerical. The induction vanishes on the axis *at the disk plane* and at both infinities, since R and R_h enter symmetrically in those limits. Upstream it is positive but weaker than the equivalent solid disk's, the bore shielding the axis. And downstream inside the bore it is *negative*: for $s > 0$ the inner term dominates because $R_h < R$, so the flow on the axis behind an annular disk is retarded. That is the centerbody wake, a real feature of annular jets, and the model produces it without being told.

Swirl is deliberately absent from Eqs. (3) and (7). The root vortex and the wake's axial vorticity lie downstream of the disk and contribute nothing ahead of it, which is the region this model exists to serve.

C. Verification

The discretized sheet is checked against Eq. (3) at stations from four radii upstream to ten downstream, holding to 2% of v_i, with the transverse components vanishing on the axis by symmetry and the error falling under refinement. The annulus is checked against its own closed form, which pins the inner sheet's sign and strength, including the reversed induction inside the bore. The momentum inversion $T = 2\rho A v_i(V_\infty + v_i)$ round-trips, and forward speed reduces v_i at fixed thrust.

One further check is the one that magnitude comparisons cannot make: the induction must point *downstream* for a thrusting disk, both in the jet and ahead of it, and must reverse when the disk axis reverses. Inverting the ring circulation sense would turn the upstream acceleration this paper rests on into a deceleration while leaving every magnitude correct.

III. Application to the demonstrator

The induction is applied as an external perturbation field to the coupled wing–body–duct solve of the companion study,[3] and the separation march is run with the fan on and off at otherwise identical conditions. The coupling is presently *one-way*: the airframe sees the fan, the fan does not see the airframe, so every result below is a lower bound on the interaction.

Table 1. Chordwise rise in axial velocity across the wing, as a percentage of freestream, at four operating points spanning the transition's range of velocity ratio $\mu = V_\infty/v_h$, from near-hover to cruise. These are a parametric sweep in μ, not trim points on a computed trajectory: no transition operating line exists yet (Sec. IV). v_h is referred to the *annular* disk area, which excludes the blade root at $r/R = 0.42$; using the full disk instead lowers μ by 8%.

V_∞	T	v_i	μ	$\Delta u_x/V_\infty$ [%] at $2y/b =$			
[m/s]	[N]	[m/s]		0.10	0.20	0.30	0.60
5	25	17.1	0.26	65.7	40.4	13.9	-1.7
12	25	14.3	0.62	23.0	14.1	4.8	-0.6
18	20	10.6	1.04	11.3	6.9	2.4	-0.3
25	12	5.9	1.86	4.5	2.8	1.0	-0.1

A. Magnitude and distribution

The quantity that matters is not the induction at a point but its variation *along the chord*, because that variation is the pressure gradient. The wing trailing edge sits roughly three times closer to the disk than its leading edge, so the induction rises substantially across the chord: at the mid-range condition $V_\infty = 12$ m/s, $T = 25$ N, the rise is 23% of freestream at $2y/b = 0.10$, 14.5% at 0.20 and 5% at 0.30. It is a favourable gradient, and it is strongest at the trailing edge, which is where turbulent separation begins and from which it marches forward.

Two features of the spanwise distribution deserve emphasis, and Fig. 4 and Table 1 carry both. It falls by a factor of four between $2y/b = 0.10$ and 0.30, concentrating the benefit inboard — over the stations that separate first.

And beyond $2y/b \approx 0.4$ it *reverses*. That sign change is not numerical, and Eq. (3) explains it: a semi-infinite vortex cylinder induces flow along its axis *inside* the streamtube and against it outside, since the total flux through any plane must be conserved and the disk adds none. A wing spanning both regions is therefore helped inboard and penalized outboard. The penalty is under 2% of freestream, an order of magnitude smaller than the inboard benefit, and it is a real property of the configuration that exists in the answer only because the upstream field is modelled at all.

B. How much incidence the fan buys

Measuring the benefit requires care, and the obvious measurement fails. Defining separation onset as a threshold crossing — the first α at which any station separates forward of a fixed chordwise station — and evaluating it on the two-degree grid used for the coefficient tables returns *no change at all* between fan on and fan off. That result is an artifact: a shift smaller than the grid spacing moves the crossing within an interval without moving it across

Table 2. Incidence bought by the fan, as the horizontal shift of the separation curve at several criterion levels. $V_\infty = 12$ **m/s,** $T = 25$ **N.**

x_{sep}/c_n	α off [deg]	α on [deg]	$\Delta\alpha$ [deg]
0.90	9.63	10.58	+0.94
0.85	11.67	12.54	+0.88
0.80	13.88	14.72	+0.85
0.75	15.44	16.20	+0.76
0.70	16.56	17.29	+0.73
0.60	18.44	19.10	+0.66
		mean, 0.90 to 0.60	+0.80

one, so the reported onset is quantized to the grid and any sub-degree shift is invisible by construction.

The appropriate measurement is the horizontal shift of the separation curve $x_{\text{sep}}(\alpha)$, evaluated at several criterion levels — Fig. 5 and Table 2. Measured that way on a 0.5° grid, the fan buys $\Delta\alpha = +0.80°$ averaged over criteria from $x_{\text{sep}} = 0.90$ to 0.60, with individual values between +0.66° and +0.94°.

The consistency across criterion levels is what makes this a property of the flow rather than of the definition: a spurious shift arising from the choice of threshold would vary strongly with it. The modest variation that remains is monotone — largest near onset (+0.94° at $x_{\text{sep}} = 0.90$), decaying as the criterion deepens (+0.66° at 0.60) — consistent with the deeper criteria crossing at higher incidence, where the same induction competes with a steeper adverse recovery.

The benefit scales as expected with the velocity ratio: at cruise speed the same comparison returns roughly a third of the transition value, consistent with the induction scaling as v_i/V_∞.

C. The same effect measured across the envelope

The measurement above is a horizontal shift of the separation curve at one operating point, on a fine incidence grid. It answers "how much incidence does the fan buy" precisely, and says nothing about what happens once the wing is past stall, where the curve it shifts no longer exists in the same form. A second measurement, from a different driver and a different data path, covers that range.

The coupled post-stall solve carries a separation point on every strip, so the powered and power-off locations can be compared outright at any attitude the solver will run. Doing that over 125 conditions — α from −4° to 90°, thrust coefficient T_c from 0.5 to 8 — gives Fig. 6 and Table 3. This is a *vertical* shift, the chord fraction held attached, where Fig. 5

Table 3. Span-mean separation point, fan off and on, at $T_c = 8$. Rows are marked according to whether the coupled solve converged or returned a limit-cycle mean, because the two are not the same kind of number.

α [deg]	$\bar{f}$ off	$\bar{f}$ on	$\Delta\bar{f}$	
-4	0.9880	0.9880	+0.0000	converged
8	0.9581	0.9611	+0.0029	converged
14	0.8811	0.8946	+0.0135	converged
16	0.8413	0.8591	+0.0179	converged
18	0.7823	0.8071	+0.0247	converged
20	0.7012	0.7414	+0.0402	cycle mean
26	0.4890	0.5450	+0.0560	cycle mean
30	0.2771	0.3420	+0.0649	cycle mean
40	0.1141	0.1511	+0.0371	cycle mean
60	0.0180	0.0247	+0.0067	cycle mean
90	0.0622	0.0171	-0.0450	cycle mean

is a horizontal one. The two are corroboration rather than duplication: they share no code beyond the geometry, and neither subsumes the other.

Three features are worth drawing out. The delay grows to a peak of +0.065 chord near $\alpha = 30°$ and decays after it, which is what a delay must do once there is little attached flow left to preserve. It is monotone in thrust at every attitude up to 70°, and nothing in the formulation enforces that: the section tables are functions of $(\eta, \alpha_{\text{local}})$ alone and contain no representation of the fan, so the delay reaches them only through the local incidence the induction field changes. And beyond 70° the sign reverses — the fan advances separation rather than delaying it — because with the plate broadside the axial induction is perpendicular to the free stream and there is no attached layer left to energise.

Two properties of the metric are stated here because both would otherwise read as physics. The *worst* strip's separation point saturates at the leading edge from $\alpha = 20°$ upward at every thrust setting, so differences taken on it past that incidence measure the metric running out of range rather than the fan ceasing to act; the span mean is quoted for that reason. And ratios of the separation shift are not sensitivities: $f(\alpha)$ is steep at the knee, so a span mean of it amplifies a smooth input, and the same solve that moves $\bar{f}$ by a factor of four across one thrust step moves the normal force by a flat 1.50–1.65 per thrust doubling at every incidence. Sensitivity should be read from the load.

IV. Limitations

The coupling is one-way, so the reported shift is a lower bound. The induction model assumes uniform disk loading; real loading tapers at both edges, softening the sheet into a band that superposed cylinders at intermediate radii would represent. The fuselage boundary layer that an aft fan genuinely ingests is not modelled, and it acts to reduce the benefit. The analysis is quasi-steady, while transition is a manoeuvre rather than a sequence of trim points. And the separation criterion inherits the companion study's caveat: bubble bursting[16] is not modelled, so the separation boundary is an upper bound on usable incidence rather than a predicted stall.

The operating points of Table 1 span the transition's range of velocity ratio but are not a trajectory through it. Confirming that they bracket a *real* transition would require checking the thrust at each speed against a trimmed flight condition, and that check cannot be made here: no mass is stated anywhere in the geometry contract, which carries shape alone. What can be confirmed is internal consistency, and it holds — μ recomputed from the contract's own disk geometry reproduces the tabulated values to within one percent at all four points. The thrust schedule also falls monotonically with airspeed, which is the qualitative shape a transition must have. Neither of those makes the set a trim schedule, and the caption no longer implies otherwise.

Two limitations attach specifically to the post-stall map of Sec. C, and not to the fine-grid measurement that precedes it. Beyond roughly $\alpha = 16^\circ$ the coupled solve does not reach a steady state; it enters a limit cycle whose mean is reported, and the separation point is evaluated on a final sweep at that mean circulation. Since f is nonlinear, $f(\bar{\gamma})$ is not $\overline{f(\gamma)}$. That gap is bounded here rather than assumed small: the leading term is second order in the cycle width, and with the per-strip incidence variance carried through the solve it evaluates to 10^{-15} to 10^{-10} against f of order unity. The cycle is wide in circulation and very nearly stationary in the local incidence f is posed in.

The second concerns the incidence sweep, which is a continuation: each solve is warm-started from the preceding attitude, so the sequence of attitudes visited forms part of the specification of a post-stall condition rather than merely the order of computation. Whether that matters was settled by repeating the entire sweep on a 1° grid and comparing the 125 conditions the two share.

It mostly does not. 113 of the 125 agree to better than 10^{-6} — the map is grid-converged over ninety percent of its extent, including every condition underlying the comparison in Sec. C: the rows at 14° and 16° agree to between 10^{-9} and 10^{-7}, and the step across them is 3.39 against 1.09 on the finer grid where it was 3.38 against 1.09 on the coarser.

The twelve that disagree are listed in Table 4, and they are isolated bistable conditions

Table 4. Conditions at which the $2°$ map and a $1°$ repeat of the same sweep disagree by more than 10^{-6}, with the solve responsible. All are past the stall.

α [deg]	T_c	solve that moved	shift	$\Delta(\Delta\bar{f})$
20	0.5	power-off	+0.0108	-0.0108
20	1	power-off	+0.0108	-0.0108
20	2	power-off	+0.0108	-0.0108
20	4	power-off	+0.0108	-0.0108
20	8	power-off	+0.0108	-0.0108
24	0.5	both	+0.0082	+0.0005
24	1	both	+0.0082	+0.0010
24	2	power-off	+0.0082	-0.0082
24	4	power-off	+0.0082	-0.0082
24	8	power-off	+0.0082	-0.0082
26	2	powered	+0.0116	+0.0116
30	2	powered	-0.0267	-0.0267

113 of 125 shared conditions agree to 10^{-6}.

rather than a drift. All lie past the stall; each is a single solve settling into a different limit cycle depending on the attitude it was approached from. Two features of that list are worth stating because they bear on how the increments should be read. At $\alpha = 20°$ only the *power-off* solve moves, so its shift of 0.0108 passes through every thrust column undiminished. At $\alpha = 24°$, $T_c \leq 1$, powered and power-off move together and the difference very nearly cancels, to 0.0005 and 0.0010. Cancellation in the increment is therefore real but not reliable, and cannot be assumed in either direction. The largest single disagreement is 0.0267 at $\alpha = 30°$, $T_c = 2$, against a fan effect of 0.0436 at the same condition.

The practical reading is that the tabulated increments are sound except at the listed attitudes, where an uncertainty of order 0.01 in $\bar{f}$ — and 0.027 in the worst case — should be carried.

V. Conclusions

[Draft: conclusions pending the two-way coupling and the transition operating line. See the repository README.]

Acknowledgments

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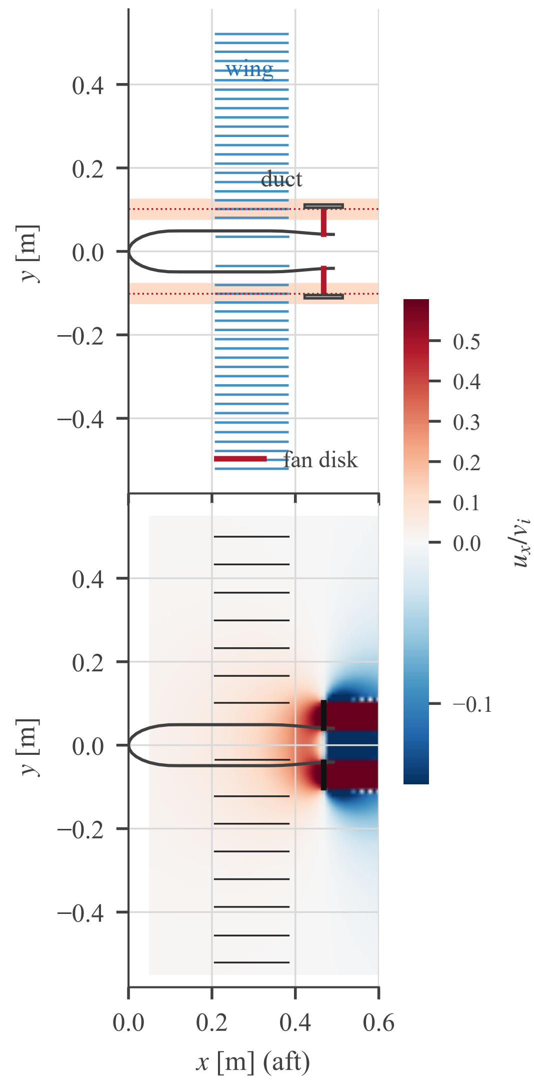

Figure 2. The coupled configuration being solved, in section. Top: planform, showing the wing lattice (44 strips, one Weissinger row each), the fuselage outline, the duct annulus and the fan disk edge-on. The shaded bands mark the span stations that separate first; the dotted lines are the fan tip radius, and their coincidence is the geometric claim of Sec. efsec:model. Bottom: the same geometry with the axial induction u_x/v_i the fan puts on it, at $V_\infty = 12$ m/s, $T = 25$ N. The wing lies in the accelerating (red) region ahead of the disk; outside the streamtube the induction reverses.

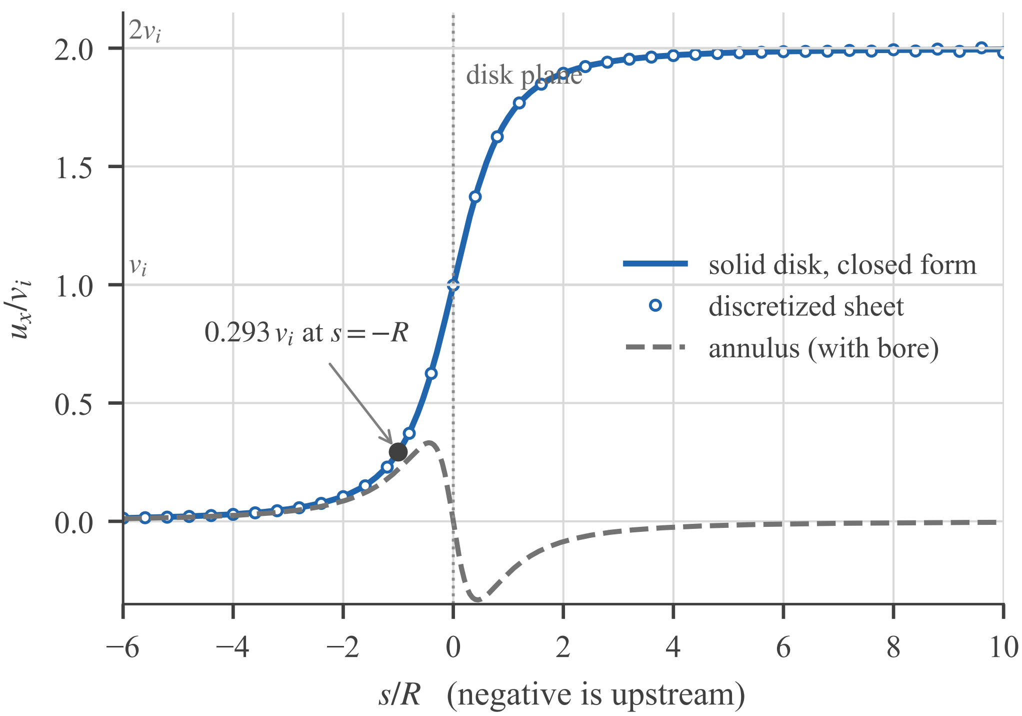

Figure 3. Axial induction on the disk axis. The discretized sheet (circles) against the closed form of Eq. (3) (solid), with the annulus (dashed) showing the reversal inside the bore that Eq. (7) predicts. Negative s is upstream — the half of the field a wake-only model does not carry, and the half this paper depends on.

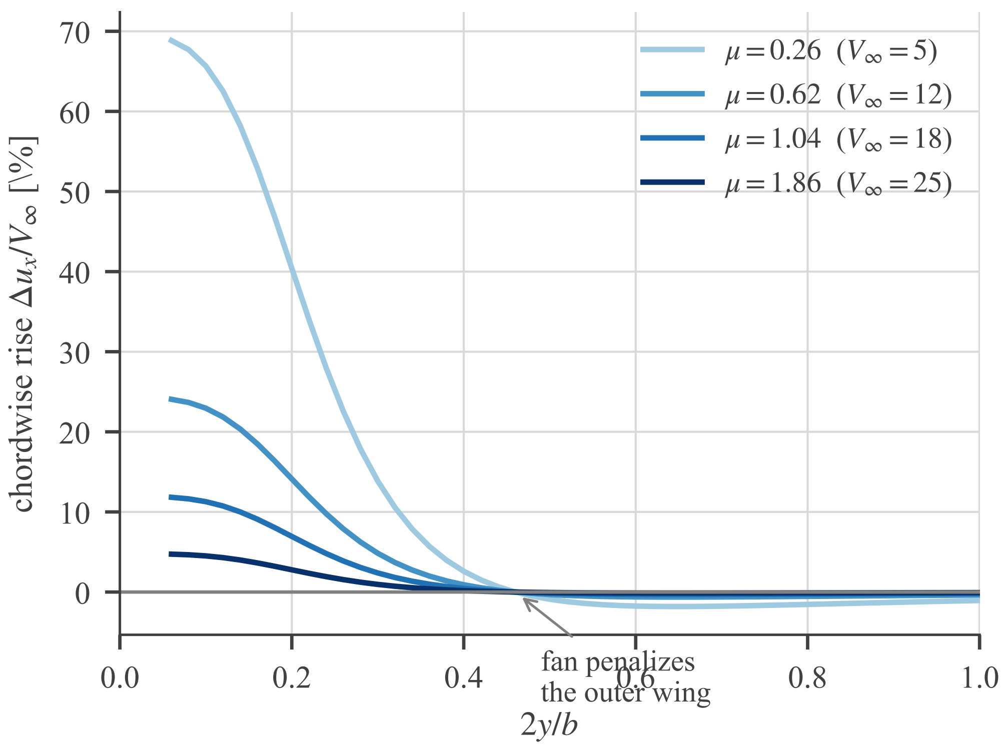

Figure 4. The chordwise rise across the span. The benefit is concentrated inboard — over the stations that separate first — and reverses beyond $2y/b \approx 0.4$, where the wing lies outside the fan's streamtube and meets its return flow.

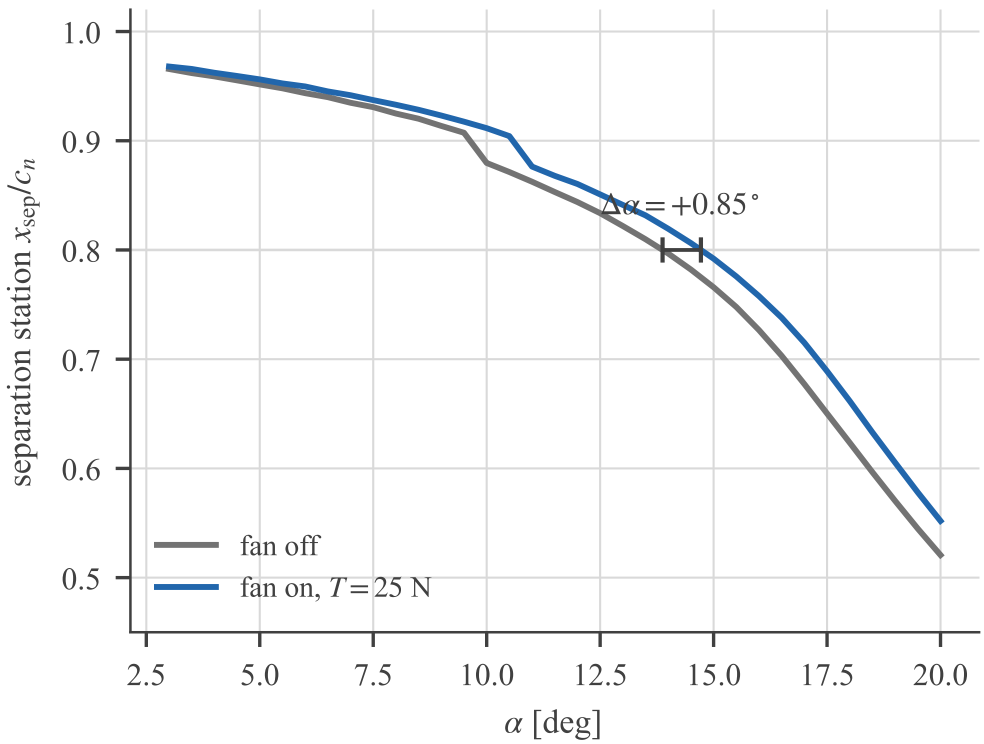

Figure 5. Separation station against incidence, fan off and on, at $V_\infty = 12$ m/s. The deliverable is the *horizontal* distance between the curves, marked at one criterion level; measuring instead where either curve crosses a fixed threshold, on a two-degree grid, returns no effect at all.

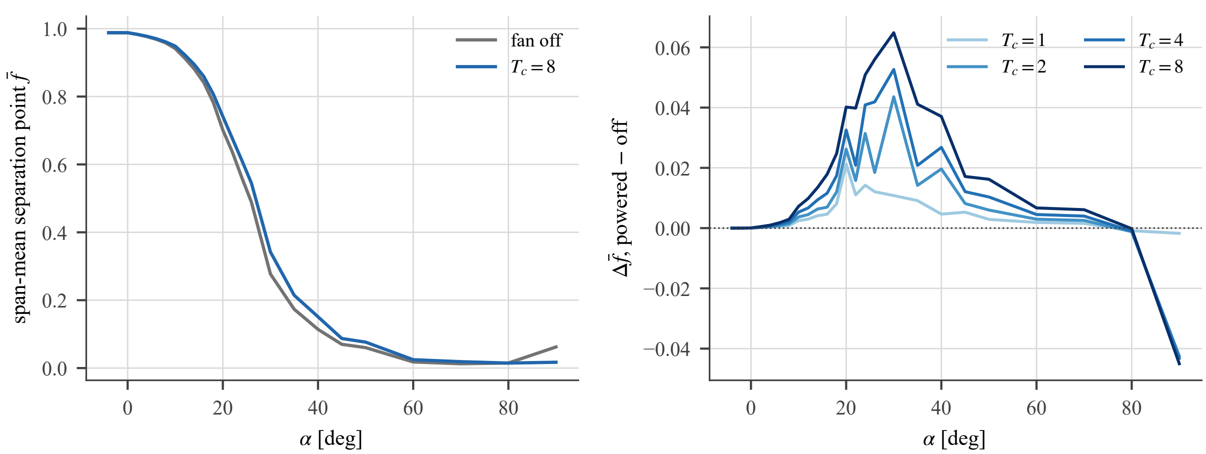

Figure 6. Span-mean separation point read directly off the coupled solve. Left: absolute location, fan off against the strongest thrust — the delay is a small displacement of a curve that is itself collapsing. Right: the delay alone, one line per thrust. The dotted zero line is drawn because the sign reverses beyond 70°.